

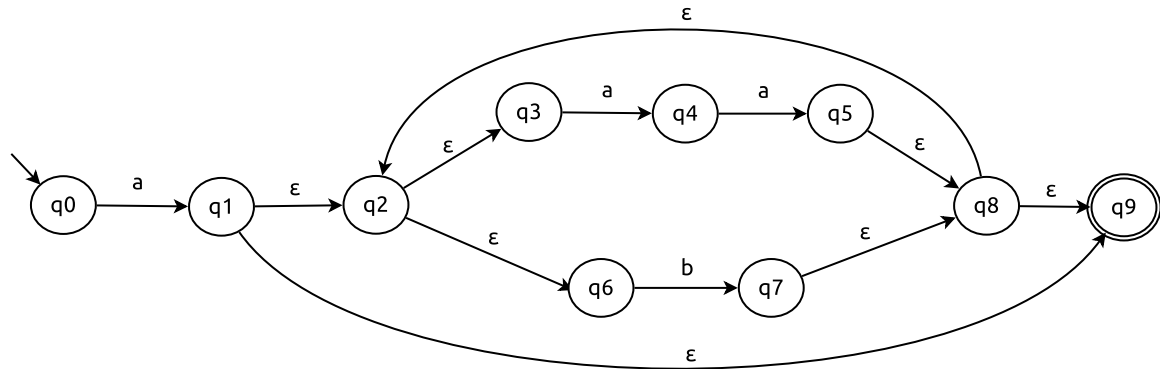
Automata Compilers

Consider the following regular expression: $a(aa \mid b)^*$.

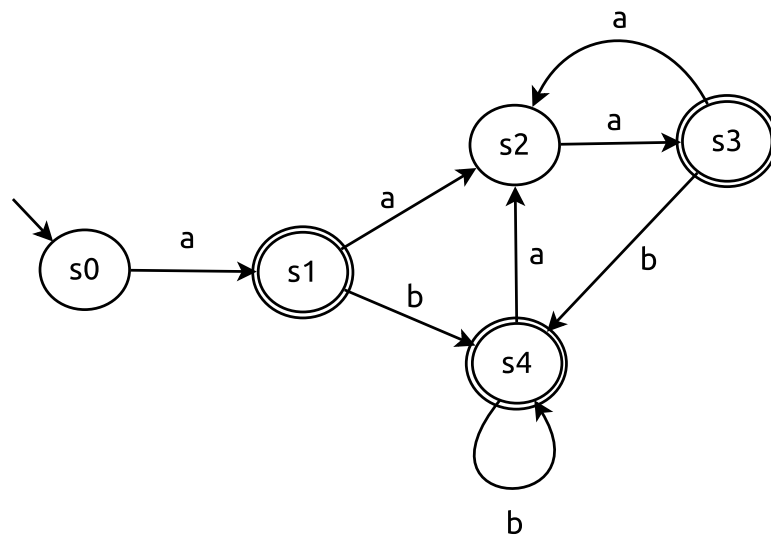
1. What language does this regular expression capture?
2. Convert this regular expression to an NFA.
3. Convert your NFA to a DFA.

Solution

1. Language: $\{a, aaa, ab, aaaaa, aaab, \dots\}$
2. NFA:



3. DFA:



where:

- $s_0 = \{q_0\}$
- $s_1 = \{q_1, q_2, q_3, q_6, q_9\}$
- $s_2 = \{q_4\}$
- $s_3 = \{q_5, q_6, q_9, q_8, q_3, q_2\}$
- $s_4 = \{q_7, q_8, q_9, q_2, q_3, q_6\}$